

In-text feedback for group 7.1 by Daniel Tyukov

Paper: "Research Review - Reconfigurable Intelligent Surfaces"

Authors: J.H. Schimmel, T.J. Geschiere, G. Gadgil

Page 1: Title Page

Title page layout

The title and authors sit on their own page with the rest left blank. In IEEE format the title, authors, abstract, and body text all start on the same first page. Having a separate title page wastes space and does not match the required format.

Suggestion: Switch to the IEEEtran LaTeX document class (`\documentclass[conference]{IEEEtran}`) or the IEEE Word template. That will put everything on the first page automatically.

Author block: "J.H. Schimmel, T.J. Geschiere, G. Gadgil"

Student numbers are not listed. The assignment requires them next to each author name.

Suggestion: Add student numbers in parentheses after each name, for example "J.H. Schimmel (1234567)."

Missing footnote

The first page needs a footnote saying: "This article was written as part of the course EE4C01 Profile Orientation & Academic Skills of the faculty Electrical Engineering, Mathematics and Computer Science of Delft University of Technology. Total word count: [fill in]." This is a requirement for the assignment.

Suggestion: In LaTeX you can add this with `\thanks{}` on the first author, or as a manual footnote.

Page 2: Research Question, Structure, AI Statement

Sections "Research question," "Structure," and "Statement on the use of generative AI"

These three sections look like they were carried over from Assignment 1 (the research proposal). They should not appear as separate sections in the actual literature review. The research question is already introduced in the introduction. The structure overview should be a sentence or two at the end of the introduction. The AI statement belongs in an Acknowledgement section at the end of the paper.

Suggestion: Remove all three sections from page 2. Work the structure overview into the introduction and move the AI statement to an Acknowledgement section after the conclusion.

Pages 2-3: Section 1 (Introduction)

Introduction, paragraph 2 (starting "An RIS consists of...")

This paragraph is clear and accessible. The explanation of how RIS works (manipulating phase, amplitude, polarization) and the distinction between passive and active configurations is well done for a general EE audience. Good work here.

Introduction, paragraph 3 (starting "Beyond the passive-active distinction...")

This paragraph brings up hybrid RIS, RF energy harvesting, IoT, and beamforming, but none of these claims have citations. If these topics are important enough to mention in the introduction, they need references to back them up. Otherwise they come across as unsupported.

Suggestion: Either add citations for each claim or remove the sentences and keep the introduction more tightly focused on the passive vs. active comparison that your paper is really about.

Introduction, paragraph 4 (starting "Experimental and simulation-based studies...")

The motivation for the review ("comparative analyses synthesizing how architectural differences fundamentally affect communication link strength remain limited") is good and clearly stated. However, it could have more impact if it came right after the paragraph about passive vs. active RIS. The paragraph in between about hybrid RIS and energy harvesting breaks the momentum a bit.

Introduction: no structure overview

The introduction does not tell the reader how the paper is organized. Readers generally expect this at the end of the introduction so they know what is coming.

Suggestion: Add a sentence at the end like: "Section II compares the simulation methods used across studies, Section III presents and compares their reported results, and Section IV discusses implications and future directions."

Page 3: Section 2 (Comparison of Methods)

Section title: "Comparison of Methods"

This section compares the methods used in the reviewed papers, which is useful. But the title could confuse readers (and the grader), because they will expect this to be the methodology section for your literature review itself. These are two different things.

Suggestion: Rename this to something like "Comparison of Simulation Setups" or "Experimental and Simulation Approaches." Then add a separate short section before it titled "Methodology" where you describe your own literature search process.

Paragraph 1 (starting "Radpour et al. simulated an RIS-panel...")

This paragraph goes straight into describing one specific study without telling the reader what the paragraph is about first. The reader does not know why this information is being presented until halfway through.

Suggestion: Start with a framing sentence, for example: "Several studies have used simulation to evaluate passive and active RIS performance in indoor scenarios, though their setups differ in important ways."

Paragraph 2 (starting "These two setups have two limitations...")

Good critical thinking here. Pointing out that both setups assume a completely blocked direct link (which inflates relative RIS performance) is an important observation. The contrast with Enahoro et al., who include the direct link, is well drawn.

Pages 4-5: Section 3 (Comparison of Results)

Section 3.1, paragraph 1 (starting "Zhang et al. [4] introduced...")

There is a typo here: "in addition to proving control over the phase" should be "providing control."

Section 3.1, paragraph 2 (starting "The sum rate was analyzed...")

Good use of concrete numbers. The comparison of sum-rate gains (1000% for active vs. 363% for passive) makes the argument tangible and easy to follow.

Section 3.2 heading: "Benifits of Adaptive Beamforming"

Typo: "Benifits" should be "Benefits."

Section 3.2 (starting "The effectiveness of a passive RIS...")

The point about beamforming algorithms being just as important as the hardware is a good observation. It adds depth beyond simply comparing passive and active configurations. The finding that reducing training patterns from 64 to 16 caused only about 1 dB SNR loss is a useful detail.

Section 3.2, last sentence: "This implies that that beamforming must be highly precise..."

There is a duplicate word here: "that that." Remove one of them.

Page 5: Sections 3.3-3.4 and Section 4

Section 3.3, paragraph 2 (starting "Enahoro et al. [3] expanded...")

Grammar issue: "Their simulations shows" should be "Their simulations show."

Section 3.4, Table 3

This summary table is one of the best parts of the paper. It gives a quick and clear comparison of the key metrics across all four sources. Well done.

Concluding paragraph after Table 3 (starting "In conclusion, the evidence from these studies suggest...")

Two things here. First, "the evidence ... suggest" should be "suggests" (subject-verb agreement). Second, this paragraph reads like a conclusion but it sits before Section 4 (Conclusion and Discussion). It feels like the paper concludes twice, which is confusing.

Suggestion: Move this paragraph into Section 4 so there is only one conclusion. Remove it from the results section.

Section 4: "Conclusion and Discussion"

This section heading appears at the bottom of page 5 but there is no content visible in the submitted draft. This obviously needs to be written for the final paper. Also, the usual convention is to put discussion before conclusion (e.g., "Discussion and Conclusion"), since you reflect on your findings first and then wrap up.

Suggestion: Write the conclusion. It should restate the research question, summarize the key findings, mention the limitations of your review, and give recommendations for future research.

Page 6: References

Reference list: only 5 sources

The assignment requires at least 9 scientific sources for a group of 3. This is the most important gap to address before the final submission.

Suggestion: Search IEEE Xplore, Google Scholar, or Scopus for more papers. You could look at topics like RIS channel estimation, RIS deployment strategies, comparisons of RIS with relay-based systems, or RIS hardware prototypes.

Reference format

The references do not follow IEEE style. For example, IEEE would format [2] as:

M. M. Amri, N. M. Tran, and K. W. Choi, "Reconfigurable intelligent surface-aided ...," IEEE Access, vol. 9, pp. 147442-147457, 2021.

The current format uses "In:" before journal and conference names, which is not standard IEEE.

Suggestion: Use the IEEEtran BibTeX style (`\bibliographystyle{IEEEtran}`) if you are using LaTeX. That handles the formatting automatically.

General Observations

Missing abstract

The paper has no abstract. The assignment requires one of 150-250 words that covers motivation, objective, method, and conclusions. The abstract is also listed as a required element, so missing it could count against you.

Suggestion: Write the abstract last, after everything else is done. Include why RIS matters for 6G, what your research question is, how you conducted the review, and the main finding (active RIS outperforms passive in link strength but at higher cost and complexity).

Word count and overall length

The paper seems quite short (about 5 pages of content). The target is 2000 words, plus or minus 10%. Adding the missing sections (abstract, methodology, more sources in the body, a proper conclusion) should bring you closer to the target.